Discrete Fourier Transform

Digital Signal Processing

The Fourier Family: Choosing the Right Tool

Signal Type	Continuous-Time	Discrete-Time
Periodic	Fourier Series (FS)	**DFT**
Aperiodic	Fourier Transform (FT)	DTFT

Core Principle

- Any signal can be expressed as a **weighted sum of sinusoids**
- The right Fourier method depends on whether the signal is **periodic/aperiodic** and **continuous/discrete**
- Periodicity in one domain $\longleftrightarrow$ Discreteness in the other

Our focus for the rest of this lecture is the bottom-right entry: the **Discrete Fourier Transform**, the tool for signals that are both sampled in time and periodic.

DFT: Mathematical Definition

For a discrete signal $x[n]$ with N samples ($n = 0, 1, \ldots, N-1$):

Analysis Equation (DFT)

$$\boxed{X[k] = \sum_{n=0}^{N-1} x[n]\, e^{-j2\pi kn/N}}, \qquad k = 0, 1, \ldots, N-1$$

Splitting into real and imaginary components:

Real Part — ReX

$$\mathrm{Re}X[k] = \sum_{n=0}^{N-1} x[n] \cos\left(\frac{2\pi kn}{N}\right)$$

Imaginary Part — ImX

$$\mathrm{Im}X[k] = -\sum_{n=0}^{N-1} x[n] \sin\left(\frac{2\pi kn}{N}\right)$$

Worked Example: Computing the Forward DFT

Take the 4-point signal $x[n] = \{1, 2, 3, 4\}$, so $N = 4$. Substituting into the analysis equations gives the spectrum below (only $N/2 + 1 = 3$ values are independent):

Sample Calculation: $k = 1$

$$\mathrm{Re}X[1] = 1(1)+2(0)+3(-1)+4(0) = -2$$

$$\mathrm{Im}X[1] = -\left[1(0)+2(1)+3(0)+4(-1)\right] = 2$$

Full Spectrum

k	$\mathrm{Re}X[k]$	$\mathrm{Im}X[k]$
0	10	0
1	-2	2
2	-2	0

- $\mathrm{Re}X[0] = 10$ is the **DC component** (sum of all samples); $\mathrm{Re}X[2] = -2$ is the **Nyquist component**, purely real as expected
- $k = 3$ is not computed separately — by conjugate symmetry it mirrors $k = 1$: $\mathrm{Re}X[3] = -2,\ \mathrm{Im}X[3] = -2$

DFT Output Size: N Samples $\to N + 2$ Values

For a **real-valued** N-point input signal:

- Full DFT produces N complex values, but for real signals...
- **Conjugate symmetry**: $X[k] = X^*[N-k]$
- Only $\frac{N}{2} + 1$ complex values are **unique**

$\Rightarrow \frac{N}{2} + 1$ values in $\mathrm{Re}X$

$\Rightarrow \frac{N}{2} + 1$ values in $\mathrm{Im}X$

$\Rightarrow$ Total: $2\left(\frac{N}{2} + 1\right) = \mathbf{N + 2}$ real values

Example: $N = 8$

- $\mathrm{Re}X[k]$, $k = 0, \ldots, 4$ $\to$ **5 values**
- $\mathrm{Im}X[k]$, $k = 0, \ldots, 4$ $\to$ **5 values**
- Total $= \mathbf{10} = \mathbf{8} + \mathbf{2}$ ✓

No New Information — Just a Different Packaging

It is worth pausing on something that looks strange at first: we started with N real samples in time and ended up with $N + 2$ real values in frequency. The extra two values do not carry new information — they come purely from how real-signal data happens to be packaged.

Key Insight: Nothing New Is Created

- For any real-valued input: $\mathrm{Im}X[0] = 0$ and $\mathrm{Im}X[N/2] = 0$ **always** (DC and Nyquist bins are purely real)
- Removing those 2 trivially-zero values: $(N + 2) - 2 = N$ **independent values**
- The frequency domain therefore carries **exactly the same** information as the N time-domain samples
- The $N + 2$ format is simply a **symmetric, convenient layout** for real-signal synthesis — not a creation of new information

Inverse DFT: Synthesis from ReX and ImX

Reconstruction of $x[n]$ from its spectral components:

IDFT Synthesis Equation

$$x[n] = \sum_{k=0}^{N/2} \frac{\mathrm{Re}X[k]}{K_k} \cos\left(\frac{2\pi kn}{N}\right) - \sum_{k=0}^{N/2} \frac{\mathrm{Im}X[k]}{K_k} \sin\left(\frac{2\pi kn}{N}\right)$$

Scaling Factors K_k

Position	**Index**	**Scale**
Interior frequencies	$k = 1, 2, \ldots, N/2 - 1$	$K_k = N/2$ (i.e. multiply by $2/N$)
DC component	$k = 0$	$K_k = N$ (i.e. multiply by $1/N$)
Nyquist component	$k = N/2$	$K_k = N$ (i.e. multiply by $1/N$)

Why We Scale — and Why the Endpoints Differ

The scaling factors on the previous slide are not arbitrary; they follow directly from how the forward DFT accumulates amplitude across N terms.

Why Scale at All

The DFT analysis equation sums N terms, accumulating amplitude as it goes. Without normalization, the reconstructed signal would come out N times too large. Dividing by N (or $N/2$) restores the correct amplitudes.

Why the Endpoints Are Different

- In the full N-point complex DFT, interior bins $k = 1, \ldots, N/2 - 1$ each appear **twice**: at k and at $N - k$ (a conjugate pair)
- Folding these symmetric pairs together doubles their contribution $\rightarrow$ scale factor $= 2/N$
- **DC** ($k = 0$) and **Nyquist** ($k = N/2$) have **no mirror counterpart** $\rightarrow$ they appear only **once** $\rightarrow$ scale factor $= 1/N$

Symmetry Properties of Sine and Cosine

Cosine: **Even Function**

$$\cos(-\theta) = +\cos(\theta)$$

- Symmetric about the y-axis
- $\mathrm{Re}X[k] = \mathrm{Re}X[N-k]$
- Even symmetry in frequency

Sine: **Odd Function**

$$\sin(-\theta) = -\sin(\theta)$$

- Antisymmetric about origin
- $\mathrm{Im}X[k] = -\mathrm{Im}X[N-k]$
- Odd symmetry in frequency

Conjugate Symmetry for Real Signals

$$X[k] = X^*[N-k] \implies \mathrm{Re}X[k] = \mathrm{Re}X[N-k], \quad \mathrm{Im}X[k] = -\mathrm{Im}X[N-k]$$

Why a Real Signal Needs Only Real Cosines and Sines

These symmetry properties explain something important about real-valued signals: their even part is built entirely from real cosines, and their odd part entirely from real sines — no complex sinusoids required.

By Euler's formula, $e^{j\theta} = \cos\theta + j\sin\theta$. For a real signal $x[n]$:

The Argument

- $\mathbf{ReX} = \sum x[n]\cos(2\pi kn/N)$ is **real and even** because cosine is even: $\cos\left(\frac{2\pi(N-k)n}{N}\right) = \cos\left(\frac{2\pi kn}{N}\right)$
- $\mathbf{ImX} = -\sum x[n]\sin(2\pi kn/N)$ is **real and odd** because sine is odd: $\sin\left(\frac{2\pi(N-k)n}{N}\right) = -\sin\left(\frac{2\pi kn}{N}\right)$
- The imaginary cross-terms from positive and negative frequencies **cancel exactly** due to antisymmetry
- As a result, only **real cosines** (even) and **real sines** (odd) are needed to fully describe a real signal

Which Fourier Method Fits Each Signal?

Signal Type	**Method**	**Spectrum**
Periodic + Continuous	**Fourier Series**	Discrete harmonic lines (bilateral)
Aperiodic + Continuous	Fourier Transform	Continuous
Aperiodic + Discrete	DTFT	Continuous + Periodic
Periodic + Discrete	**DFT**	Discrete + Periodic

Duality Rule of Thumb

- Periodicity in time $\longleftrightarrow$ Discrete spectrum in frequency
- Discrete sampling in time $\longleftrightarrow$ Periodicity in frequency

Example: A Periodic, Continuous Signal

Take a signal $x(t)$ that is **periodic and continuous** in time — for instance, a continuous waveform repeating every period T. This is exactly the case the Fourier Series is built for.

Why Fourier Series?

- Signal is *continuous* $\rightarrow$ use continuous-time Fourier analysis (not DFT, which requires discrete samples)
- Signal is *periodic* $\rightarrow$ spectrum must be **discrete**

Spectrum Characteristics:

- **Discrete impulses** (spectral lines) at $f_0, 2f_0, 3f_0, \ldots$ where $f_0 = 1/T$
- **Bilateral**: extends from $-\infty$ to $+\infty$
- Amplitudes set by Fourier Series coefficients c_k

Frequency Axis: Two Representations

Two ways to label the horizontal axis in the frequency domain:

Method 1: Sample Number k

- $k = 0, 1, 2, \ldots, N/2$
- Integer frequency bin index
- Ranges from 0 to $N/2$
- Simple integer labelling

Method 2: Fraction of Sampling Rate f

- $f = k/N$
- Normalized, dimensionless
- Ranges from 0 to 0.5
- Independent of f_s

Key Relationships

$$f = \frac{k}{N} \qquad f_{\mathrm{Hz}} = \frac{k}{N} \times f_s$$

Worked Example: Frequency Fraction for $N = 32$

Let's put these formulas to work. Take a 32-point time-domain signal transformed into the frequency domain, and look at the 9th sample of that frequency-domain output.

Setting Up

Given: $N = 32$. Using **0-based indexing** (standard in DSP, where the 1st sample is $k = 0$):

$$\text{9th sample in frequency domain} \;\Rightarrow\; k = 8$$

$$\boxed{f = \frac{k}{N} = \frac{8}{32} = \frac{1}{4} = 0.25}$$

- The 9th frequency bin represents **25.0**% of the sampling rate
- Example: if $f_s = 1000\,\text{Hz}$, this bin sits at $0.25 \times 1000 = 250.0\,\text{Hz}$

Note: with 1-based indexing instead, $k = 9 \Rightarrow f = 9/32 = 0.28125$.

Inverse DFT: Recovering the Time-Domain Signal

General IDFT Formula

$$x[n] = \frac{1}{N} \sum_{k=0}^{N-1} X[k]\, e^{j2\pi kn/N}$$

For real signals, synthesis using ReX and ImX only:

$$x[n] = \underbrace{\sum_{k=0}^{N/2} \frac{\mathrm{Re}X[k]}{K_k} \cos\left(\frac{2\pi kn}{N}\right)}_{\text{Even (cosine) part}} - \underbrace{\sum_{k=0}^{N/2} \frac{\mathrm{Im}X[k]}{K_k} \sin\left(\frac{2\pi kn}{N}\right)}_{\text{Odd (sine) part}}$$

Role of Each Array

- **ReX**$[k]$ sets the amplitude of the **cosine** at frequency k/N
- **ImX**$[k]$ sets the amplitude of the **sine** at frequency k/N (with a negative sign $\rightarrow$ phase shift)

Worked Example: IDFT of a Single Impulse in ReX

Consider a frequency-domain signal that contains **only one impulse** in ReX, with every other ReX value and every ImX value equal to zero.
Let $\mathrm{Re}X[k] = A\,\delta[k - k_0]$ (impulse at bin k_0) and $\mathrm{Im}X[k] = 0$ for all k.

$$\boxed{x[n] \;=\; \frac{A}{K_{k_0}} \cos\!\left(\frac{2\pi k_0 n}{N}\right)}$$

Why This Happens

- **ReX** drives cosine synthesis; **ImX = 0** means no sine components contribute
- A single impulse at k_0 activates **exactly one cosine basis function** $\rightarrow$ the output is a **pure cosine** at normalized frequency k_0/N
- **Special cases:**
  - $k_0 = 0$ (DC): output is a **constant** signal $x[n] = A/N$
  - $k_0 = N/2$ (Nyquist): output **alternates** $\pm A/N$ each sample